

Ashwin Mendhe

Python / Full Stack Engineer | FastAPI | Django | React | Spring Boot | AWS | Real-Time Systems

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Summary

Python / Full Stack Developer with 6+ years of experience across backend development, automation, OTT/media services, telecom, real-time streaming, and AI-driven environments. Strong experience building REST APIs and automation solutions using Python, FastAPI, Django, and Flask, with recent hands-on full-stack development using React, TypeScript, Java/Spring Boot, PostgreSQL, Redis, MQTT, and WebSocket. Experienced in real-time video and data streaming using FFmpeg, GStreamer, RTMP, and HLS, along with AWS cloud services including EC2, ECS, S3, Lambda, API Gateway, Cognito, and RDS. Hands-on experience integrating cloud applications with robots, drones, real-time telemetry, video analytics, and AI-based services. Adaptable engineer with a strong debugging and problem-solving background, experienced in using AI-assisted development tools to rapidly learn new technologies, build production features across unfamiliar technology stacks, and accelerate development while validating solutions through testing and production troubleshooting.

Skills

Tools and Frameworks

Nginx | Appium | FFmpeg | GStreamer | WebSocket | GitHub | Bitbucket | FastAPI | Zustand | React | Django | Flask | Django Rest Framework | monitoring Tools site24x7 | Celery | redis | RabbitMQ | Pandas | Numpy | RTMP | Apache kafka

Programming Languages

Python | SQL | HTML CSS JavaScript for Frontend | typescript | tailwindCss | zustand | java basic

Database

MongoDB | PostgreSQL | MySQL | ORM

Operating Systems

macOS | Linux

Cloud & DevOps

AWS (EC2, ECS, S3, Lambda, API Gateway, Cognito, RDS, Quicksight) | Google Cloud Platform (GCP) | Docker

AI & Automation

Gemini 2.0 Flash

Work Experience

Full Stack Software Engineer II Jan 2026 - Current
Mckinleyrice Pvt. Ltd. 📍 Remote

- Contributed to the end-to-end development of a cloud-based **robot and drone monitoring platform**, working across frontend, backend, real-time communication, video streaming, and production deployments.
- Developed and enhanced frontend features using **React, TypeScript, Zustand, Ant Design, and HLS.js**, including real-time monitoring dashboards, mission management, live streaming, historical work search, video playback, and work reports.
- Developed and maintained backend services using **Java and Spring Boot**, implementing REST APIs and business logic for mission execution, device management, livestream sessions, telemetry, playback, and historical records.
- Integrated **MQTT/EMQX, WebSocket, and Redis** for real-time robot telemetry, device health/status updates, mission state synchronization, and communication between robots, backend services, and web clients.
- Built and maintained real-time video workflows using **RTMP and HLS**, supporting live robot/drone video monitoring, stream lifecycle management, heartbeat handling, recorded playback, and AI video-analysis integration.
- Integrated cloud workflows with **Unitree GO2 robots and DJI drones**, working on device authentication, remote mission/task execution, robot status monitoring, DJI Cloud/Pilot integration, and livestream control.
- Implemented and improved **AI-assisted safety monitoring** workflows for construction environments, integrating live video streams with object-detection services and displaying detection results and safety events through the

web application.

- Diagnosed and resolved production issues across **React, Spring Boot, PostgreSQL, Redis, MQTT, WebSocket, and video-streaming services**, and performed build, deployment, API, database, and end-to-end testing using real robot and drone devices.
- Leveraged AI-assisted development workflows to accelerate debugging, code analysis, and adoption of unfamiliar technologies while validating implementations through builds, API testing, logs, database analysis, and end-to-end testing.

Python Automation Engineer

Nov 2022- Dec 2025

Mozark Pvt. Ltd.

- Developed a basic RAG (Retrieval-Augmented Generation) pipeline using **LangChain, Groq, ChromaDB, FAISS-CPU, and LLaMA 3.1 8B Instant** for efficient information retrieval and response generation.
- Developed and deploying RESTful API Design using **FastAPI** and **Django REST**, supporting a system that manages over **100+** devices and reducing device provisioning time by **80%**.
- Developed a **scalable** API integration and backend development reducing data processing time by **70%** for over **100+** users.
- Spearheaded development of a **FASTAPI** system utilizing **agile methodology**, enhancing SDLC efficiency by an estimated **80%**.
- Enhanced **data security** by implementing **JWT token-based authentication** in a microservices architecture, successfully supporting over **50 daily transactions** and improving **scalability**.
- Implemented comprehensive **Integration tests**, improving code quality , code reviews and reducing production issues or bugs by **80%** across **4** major projects.
- Transformed unstructured MongoDB data to structured MySQL, boosting data accessibility by **90%** via **Pandas** and **NumPy**.
- Leveraged **Apache Kafka** architecture for real-time data monitoring, processing **millions of data points** daily through distributed partitioning.
- Solid expertise in debugging complex software and device issues on Linux/macOS, including connectivity, login, and webcam/audio problems find root-cause analysis.
- Implemented efficient **coding** practices, reducing bugs by **80%** and saving **20** development hours weekly.
- Designed and validated **OAS** specifications, enhancing API documentation clarity and reducing integration errors by **80%**.
- Best Practices debugging processes by integrating the **Gemini 2.0 Flash AI** agent, improving efficiency by **70%**.
- Developed robust live video streaming systems leveraging **FFmpeg, RTMP, and AWS S3**, supporting **large-scale sports events** with high availability.
- Automated monitoring of 100+ servers and devices using custom Python plugins in Site24x7 increased system performance by 90%
- Developed **cross-platform SDKs** by creating **PyPI packages** and **CLI tools**, integrating Continuous Improvement/Continuous delivery CI/CD **pipelines** to streamline deployment and reduce release cycles by **100%**.
- Built Teams notification system using pymsteams and MongoDB for operational updates.
- Managing Python scripts to streamline **software development processes**, reducing manual task time by **80%**.

Socket Programmer

Feb 2022- Sep 2022

Ahamco / Svastir Services Pvt. Ltd.

- Developed and deployed CNN and ANN models using scikit-learn for pose estimation and classification, achieving **80%** accuracy and reducing processing time by **68%** for real-time analysis across **1000+** images.
- Deployed inference APIs on cloud functions AWS utilizing FastAPI/Django and WebSocket, achieving an **100%** uptime and supporting over **20** real-time data transactions per second.

Python Developer

Oct 2021- Feb 2022

Blockchain Innovation Pvt. Ltd.

- Developed backend modules for cryptocurrency analytics using Django and PostgreSQL, processing over **1k** transactions daily and improving data retrieval speed by **100%**.
- Developed and deployed **ETL data pipelines**, data modeling utilizing **pandas and numpy** and integrating **10+** real-time exchange APIs (e.g., Binance, WazirX), processing over **30 40** transactions daily and reducing data latency by **100%**.

Junior Research Fellow

Feb 2020- Mar 2021

IIT Bhilai

- Develop **3** Flask-based dashboards, improving data visualization for healthcare analytics, leading to a **100%** reduction in report generation time for over **50+** users.
- Automated analysis of **1K+** TCP/UDP/RTP packets using Python scripts, reducing manual analysis time by **90%**.
- **Researched** and analyzed multimedia pipelines, including **GStreamer** and **FFmpeg**, to optimize **video transmission** efficiency.

Projects

ROBOPILOT – Cloud-Based Robot & Drone Monitoring Platform

Jan 2026 - Sep 2026

Remote

Cloud-Based Robot & Drone

- ROBOPILOT is a cloud-based platform for remotely managing and monitoring autonomous robots and drones, providing real-time mission control, telemetry, live video streaming, AI-based safety monitoring, playback, and work history for construction-site operations.

Device Configurator & Software Version Controller

2025 - 2025

- Automated REST APIs for device and software management, reducing manual intervention by **90%** and decreasing deployment times by **100%**.

Live Monitoring System with Site24x7

2024 - 2024

- Real-time multi-device monitoring across 100+ servers.
- Implemented **shell scripting** to automate deployment pipelines, cutting deployment time by **100%** and enabling **daily** releases.

AI Debugging Agent

2024 - 2024

- Spearheaded the integration of **Gemini 2.0 Flash** to automate **complex workflows**, reducing manual processing time by **90%**.

Cryptocurrency Taxation Analysis

2021 - 2021

- Developed a **Django**-based system to process **financial data** and data modeling improving efficiency by **80%**.

Pose Estimation System

2022 - 2022

- Developed an ML model using OpenCV and CNNs for live video analytics, achieving a **60%** accuracy rate in object detection and reducing processing time by **45%** compared to previous methods.

Language

English	Hindi	Marathi
Native	Native	Native

Education

M.Tech in Electronics

2013 - 2015

SGGS, Nanded

B.E. in Electronics & Telecommunication

2008 - 2011

GHRCE, Nagpur

Diploma in Electronics & Communication

2004 - 2008

GHRCE, Nagpur